# Nabanita Sarker

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## Education

**BSc** **Bangladesh University of Engineering and Technology (BUET)** *June 2026*

Department of Electrical and Electronic Engineering (EEE)

- **CGPA: 3.74/4.00 (Final Year: 3.97/4.00)**
- Relevant Coursework: VLSI Circuits and Design, Optoelectronics, Semiconductor Processing & Fabrication, Microprocessor and Embedded Systems, Analog Integrated Circuits and Designs, Solid State Devices, Compound Semiconductor Devices, Introduction to Nanotechnology and Nano-electronics

## Research Interests

2D Materials | VLSI Design | Nanoelectronics | Semiconductor Devices | Neuromorphic Computing

## Research Experience

**LLM-Aided RTL Design and Verification** *Dhaka, Bangladesh*

Research Assistant (Topic: VLSI) *June 2026 – October 2026*

- Engaged under the Bangladesh Industry Research Development and Innovation (BIRDI) Grants initiative of the Skills for Industry Competitiveness and Innovation Program (SICIP)
- Prepared RTL designs and verification testbenches of assigned modules from design specifications
- Reviewed and verified LLM-generated RTL designs and verification testbenches
- Tools Used: SystemVerilog, Verilog, RTL simulation and verification tools
- *Advisor:* **Dr. Chowdhury Mohammad Rakin Haider**, Assistant Professor, CSE, BUET

**Enhanced Photocatalytic Hydrogen and Oxygen Evolution Activity by Two-Dimensional van der Waals Heterostructure: A First-Principles Study** *Dhaka, Bangladesh*

Undergraduate Thesis *April 2025 – June 2026*

- Investigated the electronic, optical, and photocatalytic properties of van der Waals (vDW) heterostructures using density functional theory (DFT)
- Studied the heterostructure as an excellent water-splitting photocatalyst with remarkable potential for fuel cells and energy storage applications
- Published in the **International Journal of Hydrogen Energy** (2026)
- Tools Used: Biovia Materials Studio, Quantum Espresso, VESTA
- *Advisor:* **Dr. Shaikh Asif Mahmood**, Professor, EEE, BUET

## Publications

[P] = Preprint, [J] = Journal

1. **[J] Nabanita Sarker**, Shaikh Asif Mahmood, *"First-Principles Analysis of AlSb/ZnTe van der Waals Heterostructure as a Self-Driven Photocatalyst for Complete Water Splitting"*. *International Journal of Hydrogen Energy*, vol. 286, 158021, 2026. (Impact Factor: 9.2) [DOI]
2. **[P] Nabanita Sarker**, *"Design-Point Transfer With Conformal Calibration for SRAM Reliability Sign-Off Across PVT Conditions"*. Research Square, 2026. [Preprint]

## Academic Projects

**High-Performance and Low-Power CMOS Implementation of a Password-Protected Car Parking System Using Cadence Design Flow** Report

- The system supports secure slot selection, password storage and verification, parking time measurement, charge calculation, and alarm generation for incorrect password attempts. Special

emphasis is placed on performance optimization (high speed, low latency), power minimization, and maximum area utilization during PnR.

- Tools Used: Cadence

**Development of a Color Intensity Based Approach for pH Estimation Using Optical Sensing** Report

- This project is about the design and implementation of a low-cost optical colorimetric pH sensing system based on LED–photodiode technology. The system determines pH by analyzing the change in color intensity of a pH indicator solution using optical absorption principles. An Arduino-based data acquisition and processing unit is used to convert optical signals into calibrated pH values, which are displayed on a computer interface.
- Tools Used: Arduino, MATLAB

**Common-Base RF Amplifier for High-Frequency Applications** Report

- Designed an RF Amplifier with required specifications of Frequency: 5 GHz, Gain greater than 8 dB PAE greater than 0.05, Psat greater than 10 dBm
- Tools Used: Cadence

**IoT Based Real Time Intravenous Fluid Monitoring and Flow Control System** Report

- Developed a monitoring system that continuously tracks fluid level in the IV bag, Automatically regulates flow rate or allows manual control by doctors/nurses via IoT interface, Sends alerts/notifications to a connected application in case of Low fluid level, Patient abnormalities (e.g., fever irregular pulse rate), Irregular flow rate
- Tools Used: Blynk, Autodesk Fusion360, Arduino

**8 Player Quiz Game Machine** Report

- Built a Machine that consists of a central quiz unit, multiple input devices (buttons) for each player, and a visual output display for presenting questions, answer choices, and player scores without microcontroller
- Tools Used: Proteus

**Versatile CNC Machine: Pen-plotter, Cutter, Engraver** Report

- Designed a CNC machine that not only draws but also can also be used in cutting and engraving purpose
- Tools Used: CNC shield and arduino

**Battery Charge Controller using Buck Converter with Auto Cut-off Feature** Report

- The project is about designing a battery charger with variable DC output using a Buck converter and an auto cutoff feature
- Tools Used: Proteus

**Real Time Fingerprint Recognition using Matlab** Report

- The goal of this project is to develop a complete system for real time data collection and fingerprint verification through extracting and matching minutiae
- Tools Used: MATLAB

## Technical Skills

- **Hardware:** Microcontrollers, IoT Devices, Sensors.
- **DFT Tools:** Biovia Materials Studio, Quantum Espresso, VESTA.
- **Circuit Simulation and Design:** Cadence, PSpice, LTspice, Proteus.
- **Others:** C, C++, AutoCAD, MATLAB, Autodesk Fusion 360, LaTeX

## Honors and Awards

- **University Merit Scholarship (Final 2 semesters)**
- **General Scholarship** in Secondary School Certificate (SSC) Examination
- **Talent Pool Scholarship** in Junior School Certificate (JSC) Examination
- **Perfect Attendance Certificate** Holy Cross College, recognized for flawless attendance